Pranav Singh Rajoria

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Education

Indian Institute of Information Technology and Management Gwalior **August 2024 – June 2028**
*B.Tech in Electrical and Electronics Engineering - CGPA : **8.36 / 10***

Experience

Software Engineering Intern (Backend) **August 2026 – Present**
Edvora | Multi-Tenant LMS | NestJS, Prisma, PostgreSQL *Remote*

- Isolated tenant data in a 74-table NestJS/Prisma LMS by forcing PostgreSQL Row-Level Security on every tenant table.
- Enforced deny-by-default access on 400+ REST endpoints via Better Auth guards, DB-backed sessions, and 5-role RBAC.
- Backed releases with 3,000+ Jest tests and a CI check failing any table without RLS; generated OpenAPI contracts for 4 client apps.

Machine Learning Intern **June 2026 – July 2026**
SmileCloud Family Dental | Healthcare AI | Python, MLX, Transformers *Remote*

- Shipped a production Text-to-SQL assistant on a 125M-parameter SELECT-only model at ∼1.5s end-to-end latency.
- Rejected unsafe non-SELECT SQL with 99.2% accuracy by training on 512K+ SQaLe, 5K OpenDental, and 10K refusal samples.
- Reduced on-device inference VRAM ∼45% and raised throughput ∼30% using sparse attention and TurboQuant quantization.

Software Engineering Intern **March 2026 – May 2026**
HireFT | AI-Powered Job Platform | Go, PostgreSQL, AWS S3, SQS *Remote*

- Sustained ∼350K job listings/day with a 20-worker concurrent Go scraping pipeline using batched PostgreSQL writes, S3, and SQS.
- Cut job-filter latency from ∼8s to ∼500ms across 500K+ records using PostgreSQL GIN indexes and application-level caching.
- Accelerated ATS resume generation from ∼120s to ∼15s via a 2-call Groq LLM pipeline for JD extraction and content generation.

Projects

OpsPilot | *Secure Incident-to-Fix Control Plane* *Go, gRPC, Python, pgvector, NATS JetStream, Docker, Clerk*

- Built an approval-gated pipeline that correlates production logs with repo evidence, generates cited RCAs, and opens draft PRs.
- Designed Go/gRPC services with **NATS JetStream + DB recovery** and **network-disabled, non-root Docker sandboxes**.
- Enforced **least-privilege execution** with tenant RBAC, **single-use approvals**, stale-SHA checks, and no merge/deploy permissions.

Visync | *Real-Time Collaborative Whiteboard* *Spring Boot, WebSockets (SockJS), Next.js, Konva.js, PostgreSQL*

- Built a **server-authoritative** whiteboard syncing **16 event types** (drawing, undo/redo, presence, chat) over WebSockets.
- Added **HMAC-signed handshake tokens**, **token-bucket rate limits**, and bounded executors with backpressure.
- Kept edits ordered via a **single-writer queue**; recovered rooms from **snapshots + event replay**, with **RDP** stroke simplification.

SentinelAPI | *API Security Scanner & Inspection Proxy* *Node.js, Express.js, MongoDB, Socket.IO, Jest*

- Built an **API security scanner and inspection proxy** that detects vulnerabilities and blocks policy-matched malicious requests.
- Implemented **50+ detection patterns** across injection, **IDOR/BOLA**, and leaked-secret checks.
- Mapped findings to **CWE/OWASP API Security** risks with severity-ranked remediation and real-time telemetry.

Technical Skills

Languages: Go, Java, Python, C++, SQL, TypeScript, JavaScript
CS Fundamentals: Data Structures & Algorithms, Computer Networks, Operating Systems, Concurrency, DBMS, Distributed Systems
Systems & Security: REST, gRPC, WebSockets, NATS JetStream, JWT, RBAC, Postgres, Docker sandboxing
Backend & Data: Spring Boot, NestJS, FastAPI, Node.js, PostgreSQL, Prisma, Redis, MongoDB, pgvector
Cloud & Tooling: Docker, AWS S3/SQS, MinIO, GitHub Actions, Linux, Jest
AI/ML: Transformers, RAG, MLX, ONNX

Achievements

- Solved **300+ DSA problems** across multiple platforms.
- **Winner, Infotsav Hackathon 2025** – won the Web Development track among 500+ participants from across India.

Leadership & Activities

- **Secretary (Technical)**, Student Activity Council – led a 20+ member team across 8 events, including Infotsav inter-college tech fest (800+ participants) and Mridang freshers' showcase (1,500+ attendees).
- **Organizer**, SIH 2026 Internal Hackathon – led a 15-member team running the campus SIH qualifier (500+ participants).
- **Operations Lead**, IEEE – led a 25-member team running logistics for Hacksagon 2025 (2,000+ registrations, 500+ finalists).